Fiscal Year:	FY 2024	**Task Last Updated:**	FY 02/12/2024
PI Name:	Brandizzi, Federica Ph.D.		
Project Title:	Life Beyond Earth: Effect of Space Flight on Seeds with Improved Nutritional Value		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	517-353-7872
Organization Name:	Michigan State University		
PI Address 1:	106 Plant Biology		
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PI Web Page:			
City:	East Lansing	**State:**	MI
Zip Code:	48824-1046	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
Start Date:	05/01/2019	**End Date:**	04/30/2024
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 04/30/2024 per NSSC (Ed., 2/24/23) NOTE: End date changed to 04/30/2023 per NSSC (Ed., 2/1/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K0707		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Plants are primary producers of food and oxygen on Earth, and will likewise be indispensable to the establishment of large-scale sustainable ecosystems and human survival on extraterrestrial surfaces. Owing to their small volume and minimal storage requirements, seeds are the most convenient means of plant transportation in space. However, through yet-largely unknown mechanisms, space flight affects seed quality and, consequently, the vigor of seedlings. Therefore, more research is needed to fully understand seed biology responses to space flight and, ultimately, make plants a sustainable source of oxygen and food in space. We plan to test the effect of space flight in the Exploration Mission-1 (EM-1) [now Artemis] Orion capsule on Arabidopsis thaliana seeds with enriched nutrient stores, specifically elevated levels of branched-chain amino acids (BCAAs). These essential nutrients serve as storage reserves for the seeds and seedlings. Generally, seeds with high levels of nutrients are more likely to germinate and produce healthy seedlings than seeds with low nutrient content.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Successful accomplishment of our research, which is grounded upon our experience with space flight with NASA Biological Research in Canisters (BRIC)-units and A. thaliana, will increase basic knowledge of the effects of space flight on fundamental plant metabolism and to facilitate germplasm selection for extended space flight and extraterrestrial colonization. Development of seeds with high levels of nutrients is important to provide compact and efficient nourishment to humans. Because space flight conditions could impoverish the nutrient reserves of seeds, we therefore hypothesize that seeds with increased levels of BCAAs will not only better endure space flight-associated stress and produce healthier seedlings, but could also supply a greater proportion of mission-critical dietary requirements. We expect that our results will also set important foundations for translatable principles needed to increase plant nutritional value and cultivation in space flight for the long-term colonization of extraterrestrial surfaces by humans.
Task Progress:	In Spring 2023, our Michigan State University team sent the Kennedy Space Center (KSC) team seeds to conduct an asynchronized ground control (ASGC) using the flight environment data to replicate as near possible the conditions experienced on Orion. Additionally, a synchronous ground control (SGC) for the ASGC was conducted without the flight environmental data at the same time as the ASGC. At the completion of the controls, seeds were returned to the Michigan State University for processing in late Spring 2023. Like the flight and flight SGC, upon receiving the samples from KSC, a portion of each sample was immediately flash frozen with liquid nitrogen and stored at -80 °C for future amino acid quantification. Since receiving all of samples (flight, flight SGC, ASGC, and ASGC SGC samples), the Michigan State University team has been working on establishing the impact space on seed amino acid content, physiology of BCAA-enriched seeds and assaying the vigor of plants from space-flown seeds. Preliminary results to date have shown that seeds flown to space have maintained germinability and viability, and that seeds with altered levels of valine and leucine have increased viability when returned from space compared to ground control. We expect to be able to complete the experiments and obtain publication quality results in the coming months.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Abstracts for Journals and Proceedings	Brandizzi F. "Life beyond Earth: Effect of space flight on seeds with improved nutritional value." 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. Abstracts. 39th Annual Meeting of the American Society for Gravitational and Space Research, Washington, DC, November 13-18, 2023. , Nov-2023
Significant Media Coverage	Bowling E. (Brandizzi F interview). "Michigan State University to send seeds around the moon. Dr. Federica Brandizzi interviewed to explain the science of this investigation on the Artemis I mission and its impact on advancing space biology knowledge about plants' responses to space flight." WILX, East Lansing, MI local channel 10, August 24, 2022. https://www.wilx.com/2022/08/24/michigan-state-university-send-seeds-around-moon/ , Aug-2022
Significant Media Coverage	Nelson S, Hollingham R. "NASA's Dr. Z. Dr. Federica Brandizzi interviewed to explain the science of this investigation on the Artemis I mission and its impact on advancing space biology knowledge about plants' responses to space flight." Space Boffins, social media podcast, December 31, 2022. https://www.thenakedscientists.com/podcasts/astronomy-podcasts/space-boffins/nasas-dr-z , Dec-2022
Significant Media Coverage	Salazar N. (Brandizzi F interview). "MSU seeds are back from space–and better than ever. Dr. Federica Brandizzi interviewed to explain the science of this investigation on the Artemis I mission and its impact on advancing space biology knowledge about plants' responses to space flight." WILS 6 News, East Lansing, MI local channel 6, January 26, 2024. https://www.wlns.com/news/msu-seeds-are-back-from-space-and-better-than-ever/ , Jan-2024
Significant Media Coverage	Lendzion L. (Brandizzi F interview). "MSU researchers bring home good news after sending seeds into orbit around moon. Dr. Federica Brandizzi interviewed to explain the science of this investigation on the Artemis I mission and its impact on advancing space biology knowledge about plants' responses to space flight." FOX47, East Lansing, MI local channel, January 31, 2024. https://www.fox47news.com/neighborhoods/msu-campus/msu-researchers-bring-home-good-news-after-sending-seeds-into-orbit-around-moo , Jan-2024